

1.4 States of Matter

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Medium

Time allowed: 80

Score: /61

Percentage: /100

Question 1a

Carbon and silicon are elements in Group 14.

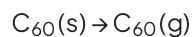
i)

Describe in simple terms the structure of solid C_{60} .

[2]

ii)

C_{60} sublimes (turns directly from solid to gas) at approximately 800 K.



Diamond also sublimes but only above 3800 K.



Explain why C_{60} and diamond sublime at such different temperatures.

[4]

[6 marks]

Question 1b

C_{60} forms hydrocarbons with similar chemical properties to those of alkenes. One such hydrocarbon is $C_{60}H_{18}$.

i)

Define hydrocarbon.

[1]

ii)

$C_{60}H_{18}$ is an alkene.

State a test to indicate the presence of double bonds between carbon atoms in alkene molecules.

[1]

iii)

State the observations seen when the test in **(b)(ii)** is carried out on an alkene.

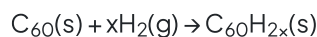
[1]

[3 marks]

Question 1c

0.144 g of C_{60} is placed in a 100 cm^3 container of hydrogen gas at a temperature of $20\text{ }^{\circ}\text{C}$ and a pressure of $1.00 \times 10^5\text{ Pa}$.

The container is heated to make the C_{60} and hydrogen gas react. The reaction occurs as shown in the equation.



After the reaction, the container is allowed to cool to $20\text{ }^{\circ}\text{C}$. The pressure decreases to $2.21 \times 10^4\text{ Pa}$. All of the C_{60} has reacted.

i)

Name the type of reaction that occurs.

[1]

ii)

Calculate the amount, in moles, of C_{60} that reacts.

amount of C_{60} = mol [1]

iii)

Calculate the amount, in moles, of hydrogen gas that reacted with the C_{60} .

Show your working.

amount of hydrogen gas = mol [3]

iv)

Use your answers from **(c)(ii)** and **(c)(iii)** to deduce the molecular formula of the hydrocarbon, $C_{60}H_{2x}$.

(If you were unable to calculate the amount of hydrogen gas, assume that 0.00240 mol of hydrogen gas reacts. This is **not** the correct value.)

Show your working.

molecular formula = [2]

[7 marks]

Question 1d

Silicon shows the same type of bonding and structure as diamond.

Silicon reacts with magnesium to form Mg_2Si .

Solid Mg_2Si reacts with dilute hydrochloric acid to form gaseous SiH_4 and a solution of magnesium chloride.

i)

Construct an equation for this reaction. Include state symbols.

[2]

ii)

Predict the shape of the SiH_4 molecule.

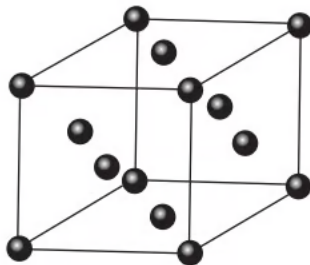
[1]

[3 marks]

Question 2a

Copper and iodine are both solids which have different physical and chemical properties.

Each element has the same face-centred crystal structure which is shown below.



The particles present in such a crystal may be atoms, molecules, anions or cations. In the diagram above, the particles present are represented by the black balls.

Which type of particles are present in the iodine crystal? Give their formula.

particle

formula

[2 marks]

Question 2b

When separate samples of copper or iodine are heated to 50 °C, the copper remains as a solid while the iodine turns into a vapour.

Explain, in terms of the forces present in the solid structure, why copper remains a solid at 50 °C.

[2 marks]

Question 2c

Explain, in terms of the forces present in the solid structure, why iodine turns into a vapour when heated to 50 °C.

[2 marks]

Question 2d

Calculate the volume, in cm^3 , that 5.00 g of iodine vapour occupies at 185°C and 100 kPa.

The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

Give your answer to 3 significant figures.

[4 marks]

Question 2e

Although copper is a relatively unreactive metal, when it is heated to a high temperature in an excess of chlorine, copper(II) chloride is formed.

i)

How does chlorine behave in this reaction?

[1]

ii)

When a mixture of copper and iodine is heated to a high temperature, no reaction occurs.

Suggest a reason for this difference.

[1]

[2 marks]

Question 3a

This question is about ideal gas theory.

State the basic assumptions of the kinetic theory as applied to an ideal gas.

[4 marks]

Question 3b

Carbon dioxide does not behave as an ideal gas.

Suggest **one** reason why not.

[1 mark]

Question 3c

Explain why CO_2 is a gas at room temperature.

[2 marks]

Question 3d

Carbon dioxide can be used to inflate life jackets.

Explain the effect of decreasing the temperature of carbon dioxide on the pressure inside a life jacket.

[3 marks]

Question 4a

Until 1985, carbon was thought to exist in only two structural forms or allotropes. In 1985 another form, buckminsterfullerene, was discovered, in which the carbon exists as spherical molecules.

The other two forms of carbon have very different structures.

i)

Name these two forms.

[1]

ii)

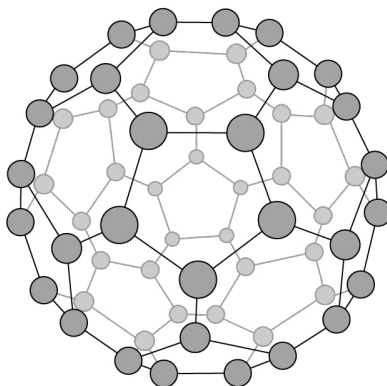
Give **three** differences in physical properties between these two forms.

[3]

[4 marks]

Question 4b

The diagram shows the structure of buckminsterfullerene.



The molecule of buckminsterfullerene contains 60 carbon atoms.

Suggest a reason why buckminsterfullerene reacts with hydrogen under suitable conditions and give a formula for the product.

[2 marks]

Question 4c

In 2010, two scientists from the University of Manchester were awarded the Nobel Prize for Physics for their work on graphene, a 'nano-material' and new structural form of carbon.

i)

Graphene is in the form of sheets of carbon one atom thick.

Calculate the number of carbon atoms present in a sheet of graphene with a mass of one thousandth of a gram (0.001 g).

[1]

ii)

The number of hexagons in a large sheet of graphene can be assumed to be one half of the number of carbon atoms. Each hexagon has an area of 690 nm^2

Calculate the area of the sheet of graphene in (i).

[2]

[3 marks]

Question 4d

Explain whether you would expect samples of graphene and buckminsterfullerene to be electrical conductors.

[2 marks]

Question 5a

The kinetic theory of gases is used to explain the large scale (macroscopic) properties of gases by considering how individual molecules behave.

State **two** basic assumptions of the kinetic theory as applied to an ideal gas.

[2 marks]

Question 5b

State **two** conditions under which the behaviour of a real gas approaches that of an ideal gas.

[2 marks]

Question 5c

Place the following gases in decreasing order of ideal behaviour.

ammonia, neon, nitrogen

most ideal least ideal

Explain your answer.

[3 marks]

Question 5d

Explain why a liquid eventually becomes a gas as the temperature is increased.

[2 marks]

Model Answers: Medium

1a

a)

i) The structure of solid C_{60} is:

- Simple molecular; [1 mark]
- Lattice / regular arrangement of C_{60} molecules; [1 mark]

ii) C_{60} and diamond sublime at such different temperatures because:

- C_{60} has weak intermolecular / VdW / London / dispersion / id-id forces and covalent bonds; [1 mark]
- Diamond has strong covalent bonds; [1 mark]
- More energy required to break; [1 mark]
- Lots of covalent bonds in diamond / (complex) 3D network / giant covalent structure; [1 mark]

[Total: 6 marks]

- C_{60} is buckminsterfullerene
 - Although it has strong covalent bonds between the atoms, it has weak intermolecular forces between the molecules
 - It is these that are broken when the substance sublimates
 - Being weak, not a lot of energy is needed to overcome these, so it sublimates at low temperatures
- Diamond, also consisting of carbon atoms, has a giant covalent structure
 - It has strong covalent bonds between carbon atoms and it is these that require breaking
 - They need much more energy to do so, compared to intermolecular forces, resulting in a much higher temperature needed for sublimation

1b

b)

i) A hydrocarbon is:

- A molecule / compound that is made up of carbon and hydrogen atoms **only**; [1 mark]

ii) A test to indicate the presence of double bonds between carbon atoms in alkene molecules is:

- Add bromine water / Br₂(aq); [1 mark]

iii) When the test in **(b)(ii)** is carried out on an alkene, the observation made is:

- (It would turn from) orange / brown to colourless; [1 mark]

[Total: 3 marks]

- For the definition of a hydrocarbon you **must** make sure you include the word 'only' in your answer
 - Failing to do so will result in the mark for the definition being lost
- Bromine water is an orange-brown liquid
- When it is added to an alkene, the bromine adds across the double bond to form a halogenoalkane resulting in the solution turning colourless
- A saturated compound such as an alkane would not react with the bromine water and would remain orange

1c

c)

i) The type of reaction that occurs is:

- Addition / hydrogenation; [1 mark]

ii) The amount, in moles, of C₆₀ that reacts is:

- $0.144 / 720 = 2.0 \times 10^{-4}$

OR

$$2.00 \times 10^{-4}; [1 \text{ mark}]$$

iii) The amount, in moles, of hydrogen gas that reacted with the C_{60} is:

- Use of $pV = nRT$; [1 mark]
- Use of $n = pV / RT$

OR

$$\Delta n = (p_1 - p_2) V / RT; [1 \text{ mark}]$$

- $\Delta n = (1.00 \times 10^5 - 2.21 \times 10^4) \times 100 \times 10^{-6} / (8.31 \times 293) = 0.0032$

OR

$$0.00320; [1 \text{ mark}]$$

iv) The molecular formula of the hydrocarbon, $C_{60}H_{2x}$ is:

- $(C_{60} : H_2) = 2.00 \times 10^{-4} : 0.00320$ or $1 : 16$; [1 mark]
- $C_{60}H_{32}$; [1 mark]

[Total: 7 marks]

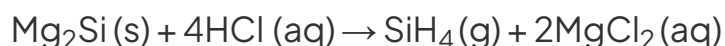
- An addition reaction is one where two (or more) molecules combine to give a single product
 - In this case, C_{60} and H_2 are combining to form the only product, $C_{60}H_{32}$
 - The addition of hydrogen to a substance is called hydrogenation, so this would also be accepted as an answer
- To calculate the number of moles of C_{60}
 - Find the molar mass of C_{60} : $12 \times 60 = 720$
 - Then use $\frac{\text{mass}}{\text{molar mass}} = \frac{0.144}{720} = 2.0 \times 10^{-4} \text{ mol}$
- To calculate the number of moles of hydrogen **which reacted**:
 - You will get one mark for stating the ideal gas equation: $pV = nRT$
 - You get a second mark for rearranging to find n (number of moles) so $n = \frac{pV}{RT}$
 - Your third mark is for completing the calculation
 - Calculate the difference between the start and end pressure: $1.00 \times 10^5 - 2.21 \times 10^4 = \mathbf{77900 \text{ Pa}}$
 - Convert the volume from cm^3 to m^3 : $100 \text{ cm}^3 \times 1000000 = \mathbf{1 \times 10^{-4} \text{ m}^3}$
 - Input your numbers into the equation: $n = \frac{77900 \times 1 \times 10^{-4}}{8.31 \times 293} = \mathbf{0.00320 \text{ mol}}$
- To deduce the molecular formula of the hydrocarbon, $C_{60}H_{2x}$:

- The moles of carbon: hydrogen as a ratio is: 0.0002: 0.00320 or 1: 16
- If x is 16 then H = (2 x 16) = 32

1d

d)

i) An equation for this reaction with state symbols is:



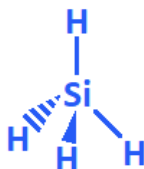
- Species **AND** balancing; [1 mark]
- State symbols; [1 mark]

ii) The shape of the SiH₄ molecule is:

- Tetrahedral; [1 mark]

[Total: 3 marks]

- You are given the formula for one reactant, Mg₂Si and one product, SiH₄
- You should know the formula of the other reactant, hydrochloric acid, HCl, and other product, magnesium chloride, MgCl₂
- Write an unbalanced equation: .
 - $\text{Mg}_2\text{Si} + \text{HCl} \rightarrow \text{SiH}_4 + \text{MgCl}_2$
- To balance the equation:
 - Balance the magnesium atoms first:
 - $\text{Mg}_2\text{Si} + \text{HCl} \rightarrow \text{SiH}_4 + \mathbf{2\text{MgCl}_2}$
 - Then balance the chlorine and hydrogen atoms:
 - $\text{Mg}_2\text{Si} + \mathbf{4\text{HCl}} \rightarrow \text{SiH}_4 + \mathbf{2\text{MgCl}_2}$
- To determine state symbols, read the question:
 - **Solid** Mg₂Si reacts with **dilute** hydrochloric acid to form **gaseous** SiH₄ and a **solution** of magnesium chloride
 - From this you can deduce the four state symbols you need:
 - $\text{Mg}_2\text{Si (s)} + 4\text{HCl (aq)} \rightarrow \text{SiH}_4\text{ (g)} + 2\text{MgCl}_2\text{ (aq)}$
 - If the question tells you to include state symbols, it is likely you will lose a mark if they are not included
- The structure of SiH₄ is:



- SiH_4 is similar in shape to CH_4 as both Si and C are in Group 4 and will form 4 bonding pairs of electrons with 4 hydrogen atoms
- These four bonding pairs of electrons will repel each other equally resulting in bond angles of 109.5° and a tetrahedral shape

2a

a) The type of particles present in the iodine crystal and formula are:

- Molecules; [1 mark]
- I_2 ; [1 mark]

[Total: 2 marks]

- A molecule can be defined as a group of atoms
 - Group 7 elements such as iodine exist as molecules because two atoms share their electrons to achieve a full outer shell resulting in the formation of molecules with the formula I_2 , Cl_2 and so on

2b

b) Copper remains a solid at 50°C because:

- Cations are held in a 'sea' of delocalised electrons; [1 mark]
- By strong metallic bonds; [1 mark]

[Total: 2 marks]

- Cations are positively charged ions
- The metallic bond exists between these and the delocalised electrons that are free to move throughout the structure
- Lots of energy is required to overcome these strong bonds hence the high melting point

2c

c) Iodine turns into a vapour when heated to 50°C because:

- There are instantaneous dipole- induced dipole forces between molecules
OR
London dispersion forces; [1 mark]
- These forces are weak (so require little energy to overcome them); [1 mark]

[Total: 2 marks]

- Make sure you are specific about the type of force that is being overcome
 - You cannot state that van der Waals' forces are overcome as the specification uses this as a generic term to describe all intermolecular forces

2d

d) The volume, in cm³, that 5.00 g of iodine vapour occupies at 185 °C and 100 kPa is:

- $n = \frac{5}{253.8} = 0.0197 \text{ mol}$; [1 mark]
- $T = 458 \text{ K}$ **AND** $P = 100\,000 \text{ Pa}$; [1 mark]
- $V = \frac{nRT}{P}$ **OR** $\frac{0.0197 \times 8.31 \times 458}{100\,000}$ **OR** $7.50 \times 10^{-4} \text{ (m}^3\text{)}$; [1 mark]
- $V = 750 \text{ (cm}^3\text{)}$; [1 mark]

[Total: 4 marks]

- The number of moles is found using $\frac{\text{mass}}{\text{molar mass}}$
- The ideal gas equation is $PV = nRT$ so this needs rearranging to find V
- Units for temperature should be in Kelvin and pressure in Pascals
 - To convert from °C to K, add 273
 - To convert from kPa to Pa, multiply by 1000
- Units for volume will be in m³ so this needs converting to cm³ by multiplying by 10⁶

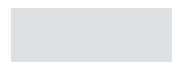
2e

e)

i) Chlorine behaves as an:

- Oxidising agent; [1 mark]

ii) No reaction occurs when a mixture of copper and iodine is heated to a high temperature because:



i) Chlorine behaves as an:

- Oxidising agent; [1 mark]

ii) No reaction occurs when a mixture of copper and iodine is heated to a high temperature because:

- Iodine is a weaker oxidising agent than chlorine; [1 mark]

[Total: 2 marks]

- An oxidising agent is one which oxidises another element, causing it to lose electrons
 - The oxidation number of the substance being oxidised will increase
 - Copper in its elemental state has an oxidation of 0
 - The oxidation number of copper in copper chloride is +2
- An oxidising agent will itself be reduced, gaining electrons
 - The oxidation number of the oxidising agent decreases
 - Chlorine in its elemental state has an oxidation number of 0
 - The oxidation number of chlorine in copper chloride is -2
- The command word suggest means that you may not have come across this topic specifically but that you should apply your understanding
 - Using your knowledge from part i) you should be able to deduce that iodine is not a strong enough oxidising agent to cause copper to lose electrons

3a

a) The basic assumptions of the kinetic theory as applied to an ideal gas are:

Any **four** from:

- The molecules behave as rigid spheres; [1 mark]
- There are no / negligible intermolecular forces between the molecules; [1 mark]
- The molecules have no / negligible volume; [1 mark]
- The molecules move in random motion; [1 mark]
- The temperature of the gas is related to the average kinetic energy of the molecules; [1 mark]
- Collisions between the molecules are perfectly elastic; [1 mark]

[Total: 4 marks]

- You can assume you need to give four assumptions as the question is worth 4 marks

3b

b) Carbon dioxide does not behave as an ideal gas because:

- There are intermolecular forces between CO₂ molecules
OR
CO₂ molecules have volume; [1 mark]

[Total: 1 mark]

- Carbon dioxide molecules have London dispersion forces between them so do not exactly fit the criteria to be classed as an ideal gas but instead is classed as a real gas

3c

c) CO₂ is a gas at room temperature because:

- CO₂ has a simple molecular structure; [1 mark]
- It has instantaneous dipole-induced dipole interactions / London dispersion forces / van der Waals' forces / weak intermolecular forces; [1 mark]

[Total: 2 marks]

- Although you do not have to be specific about the type of intermolecular force that is present to achieve the mark, it is good practice to do, as other questions will require this

3d

d) The effect of decreasing the temperature of the carbon dioxide on the pressure inside the life jacket is:

- The pressure inside the life jacket decreases; [1 mark]
- (Because) the particles lose kinetic energy; [1 mark]

- There are less frequent collisions with the life jacket wall; [1 mark]

[Total: 3 marks]

- **Remember:** Pressure is generated when gas molecules hit the wall of a container
 - The more collisions there are, the higher the pressure
 - Temperature is related to the average kinetic energy of the particles
 - The lower the temperature, the less kinetic energy the particles have
 - Increasing the volume of a container also decreases the pressure, again due to fewer collisions between the gas molecules and the container wall

4a

a)

i) These two forms are called:

- Diamond
AND
Graphite; [1 mark]

ii) **Three** differences in physical properties between these two forms are:

Any **three** from:

- Graphite can conduct (electricity) but diamond can't; [1 mark]
- Graphite is soft / slippery but diamond is hard; [1 mark]
- Graphite is black but diamond is transparent / colourless; [1 mark]
- Graphite is less dense than diamond; [1 mark]
- Graphite has a lower melting point than diamond; [1 mark]

[Total: 4 marks]

- Make sure you refer to both graphite and diamond for each different physical property to achieve each mark
- **Remember:** In graphite, each carbon atom is only bonded to three other carbon atoms whereas in diamond each carbon is bonded to four
 - This leaves an electron in graphite free to move between the layers and carry a charge
 - The layers are also able to slide over each other resulting in graphite being soft

4b

b) Buckminsterfullerene reacts with hydrogen under suitable conditions and a formula for the product is:

- Each carbon is bonded to three other carbons / has three bonding locations
OR
It is unsaturated / contains double bonds; [1 mark]
- $C_{60}H_{60}$; [1 mark]

[Total: 2 marks]

- This question uses the command word suggest, meaning you might not have come across this question specifically but should apply your understanding of bonding to the situation
- You can see from the diagram, that each carbon atom is bonded to three other carbon atoms, which involves three out of the four outer electrons on each carbon atom
- This fourth electron could either form a double bond between two carbon atoms, resulting in an unsaturated molecule, or the electron could become delocalised (just like in the structure of graphite)
 - Both of these would allow the substance to react
- In reality, the structure is much more complicated but goes beyond the scope of this course
- If each carbon atom can form one bond then the number of carbons and hydrogen atoms will be equal

4c

c)

i) The number of carbon atoms present in a sheet of graphene with a mass of one thousandth of a gram is:

- $\frac{0.001 \times 6.02 \times 10^{23}}{12.0} = 5.02 \times 10^{19}$; [1 mark]

ii) The area of the sheet of graphene in (i) is:

- Number of hexagons present: $\frac{5.02 \times 10^{19}}{2} = 2.51 \times 10^{19}$; [1 mark]
- Area of sheet = $690 \times 2.51 \times 10^{19} = 1.73 \times 10^{22} \text{ nm}^2$; [1 mark]

[Total: 3 marks]

- To find the number of carbon atoms present:
 - Number of moles = $\frac{\text{mass}}{M_r} = \frac{0.001}{12} = 8.00 \times 10^{-5}$
 - Number of atoms = number of moles x Avogadro's constant = $8.00 \times 10^{-5} \times 6.02 \times 10^{23} = 5.02 \times 10^{19}$
 - You are given the value of Avogadro's constant in an exam
- You are told in the question that the number of hexagons can be assumed to be one half of the number of carbon atoms so $5.02 \times 10^{19} / 2 = 2.51 \times 10^{19}$
- This figure can be multiplied by the area of one hexagon to find the whole area of the sheet

4d

d) Explaining whether you would expect samples of graphene and buckminsterfullerene to be electrical conductors:

- Graphene would conduct because there are free / delocalised / mobile electrons; [1 mark]
- Buckminsterfullerene would not because it consists of simple molecules / particles; [1 mark]

[Total: 2 marks]

- Although the buckminsterfullerene does contain delocalised electrons, these are located within separate molecules so they cannot flow through the entire structure and carry a charge

5a

a) **Two** basic assumptions of the kinetic theory as applied to an ideal gas are:

Any **two** from:

- There are no intermolecular forces between ideal gas molecules; [1 mark]
- Ideal gas molecules have no volume; [1 mark]

- Collisions between ideal gas molecules are perfectly elastic; [1 mark]
- Ideal gas molecules behave as rigid spheres; [1 mark]
- The molecules move in random motion; [1 mark]
- The temperature of the gas is related to the average kinetic energy of the molecules; [1 mark]

[Total: 2 marks]

- As you can see, this is a common question to be asked, sometimes you need to give more than two assumptions so it is worth investing the time to learn them all

5b

b) Two conditions under which the behaviour of a real gas approaches that of an ideal gas are:

- High temperature; [1 mark]
- Low pressure; [1 mark]

[Total: 2 marks]

- At high temperatures and low pressure:
 - The particles have more kinetic energy, so move faster
 - Being at low pressure means they can move further apart from each other resulting in negligible intermolecular forces
 - An ideal gas does not have intermolecular forces

5c

c) The gases in decreasing order of ideal behaviour are:

- Neon, nitrogen, ammonia; [1 mark]
- Nitrogen has stronger instantaneous dipole-induced dipole / London dispersion forces than neon; [1 mark]
- Ammonia has hydrogen bonding as well as instantaneous dipole-induced dipole / London dispersion forces; [1 mark]

[Total: 3 marks]

- The stronger the intermolecular forces, the less the substance acts as an ideal gas

- **Remember:** No intermolecular forces is one assumption of kinetic theory
- In the specification, van der Waals' forces are used as a generic term to describe all intermolecular forces so you must be specific as to which type is found in each gas

b) Two conditions under which the behaviour of a real gas approaches that of an ideal gas are:

- High temperature; [1 mark]
- Low pressure; [1 mark]

[Total: 2 marks]

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[Total: 3 marks]

- The stronger the intermolecular forces, the less the substance acts as an ideal gas
 - **Remember:** No intermolecular forces is one assumption of kinetic theory
- In the specification, van der Waals' forces are used as a generic term to describe all intermolecular forces so you must be specific as to which type is found in each gas
- Nitrogen is a diatomic molecule consisting of more electrons than neon so there will be stronger instantaneous dipole-induced dipole / London dispersion forces
- Ammonia, NH_3 , has a nitrogen atom with a lone pair of electrons that can form a hydrogen bond with a hydrogen atom on neighbouring molecule
- Hydrogen bonds are the strongest type of intermolecular force

5d

d) A liquid eventually becomes a gas as the temperature is increased because:

- The kinetic energy of the particles / molecules increases; [1 mark]
- Intermolecular forces are more easily broken / overcome; [1 mark]

